

Association Between Bowel Movement Pattern and Cognitive Function

Prospective Cohort Study and a Metagenomic Analysis of the Gut Microbiome

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Abstract

Background and Objectives

Little is known regarding the association between intestinal motility patterns and cognitive function in individuals who are baseline cognitively healthy. The gut microbiome may contribute to the association. We examined the association between bowel movement (BM) pattern and cognitive function and explored the role of the gut microbiome in explaining this association.

Methods

In this prospective study, we leveraged 3 cohort studies, Nurses' Health Study (NHS), NHSII, and Health Professionals Follow-Up Study (HPFS). Participants reported BM frequency and subjective cognitive function. In a subset of NHSII participants, we assessed cognitive function using an objective neuropsychological battery. We profiled the gut microbiome in a subset of participants using whole-genome shotgun metagenomics. General linear models, Poisson regression, and logistic regression were used to quantify the association of BM frequency with different cognitive measurements.

Results

We followed 112,753 men and women (women: 87.6%) with a mean age of 67.2 years at baseline (NHS: 76 years, NHSII: 59 years, HPFS: 75 years) for a median follow-up of 4 years (NHSII and HPFS: 4 years, NHS: 2 years). Compared with those with BM once daily, participants with BM frequency every 3+ days had significantly worse objective cognitive function, equivalent to 3.0 (95% confidence interval [CI], 1.2–4.7) years of chronological cognitive aging. We observed similar J-shape dose-response relationships of BM frequency with the odds of subjective cognitive decline and the likelihood of having more subsequent subjective cognitive complaints (both $p_{\text{nonlinearity}} < 0.001$). BM frequencies of every 3+ days and $\geq$ twice/day, compared with once daily, were associated with the odds ratios of subjective cognitive decline of 1.73 (95% CI 1.60–1.86) and 1.37 (95% CI 1.33–1.44), respectively. BM frequency and subjective cognitive decline were significantly associated with the overall gut microbiome configuration (both $p < 0.005$) and specific microbial species in the 515 participants with microbiome data. Butyrate-producing microbial species were depleted in those with less frequent BM and worse cognition, whereas a higher abundance of proinflammatory species was associated with BM frequency of $\geq$ twice/day and worse cognition.

Discussion

Lower BM frequency was associated with worse cognitive function. The gut microbial dysbiosis may be a mechanistic link underlying the association.

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Glossary

AD = Alzheimer dementia; **APOE** = apolipoprotein E; **BM** = bowel movement; **BMI** = body mass index; **CI** = confidence interval; **HPFS** = Health Professionals Follow-Up Study; **MCI** = mild cognitive impairment; **NHS** = Nurses' Health Study; **PD** = Parkinson disease; **PERMANOVA** = permutational multivariate analysis of variance; **SES** = socioeconomic status; **sFFQ** = semiquantitative food frequency questionnaire.

Introduction

The frequency of bowel movements is affected by diet, physical activity, use of medications, and several medical conditions, including hormonal, gastrointestinal, and neurologic diseases. Among the latter, the most notable is Parkinson disease (PD), in which constipation often precedes other features by several years.^{1,2} In addition, fecal incontinence is a classic symptom of late-stage dementia.³ However, few studies have studied bowel movement frequency in relation to neurodegenerative changes beyond PD. A recent cross-sectional study reported that infrequent bowel movements were associated with a higher prevalence of mild cognitive impairment (MCI) in 751 participants in Singapore.⁴ However, little is known regarding the association between bowel movements and cognitive function in individuals who are baseline cognitively healthy.

In vitro and in vivo evidence supports the notion that the gut microbiome, through dynamic communications along the 'gut-brain axis,' contributes to the pathogenesis of Alzheimer dementia (AD) and cognitive impairment.^{5,6} In addition, gastrointestinal motility disorders are closely associated with gut dysbiosis and alterations in the gut microbiome.^{7,8} Therefore, bowel movements, gut microbiome, and crosstalk between them may contribute to age-related cognitive decline. A recent study of over 500,000 participants from the UK Biobank reported an association between laxative use and increased the risk of all-cause dementia and suggested that the gut microbiome could be a mediating factor behind this association.⁹ However, few studies have investigated this potential mechanistic link in humans using high-resolution microbial profiling technology and examined species-level microbial features. A few cross-sectional human studies reported altered microbial ecological metrics, for example, decreased richness and α -diversity, and differentially abundant microbial features at the genus level, in individuals with poor cognitive function and patients with AD.^{10,11}

We, therefore, conducted this prospective study to investigate the associations of bowel movement pattern with objectively measured cognitive function in 12,696 women in the Nurses' Health Study II (NHSII) and subjective cognitive function in 113,145 women and men from 3 cohort studies, the Nurses' Health Study (NHS), NHSII, and the Health Professionals Follow-Up Study (HPFS). We hypothesized that participants with disordered bowel movements, defined as extreme ends of bowel movement frequency, had worse cognitive function than those with daily bowel movements. In an exploratory

analysis, we examined the interrelationship among bowel movement frequency, the gut microbiome, and subjective cognitive function in a subpopulation of 515 participants from NHSII and HPFS.

Methods

Study Population

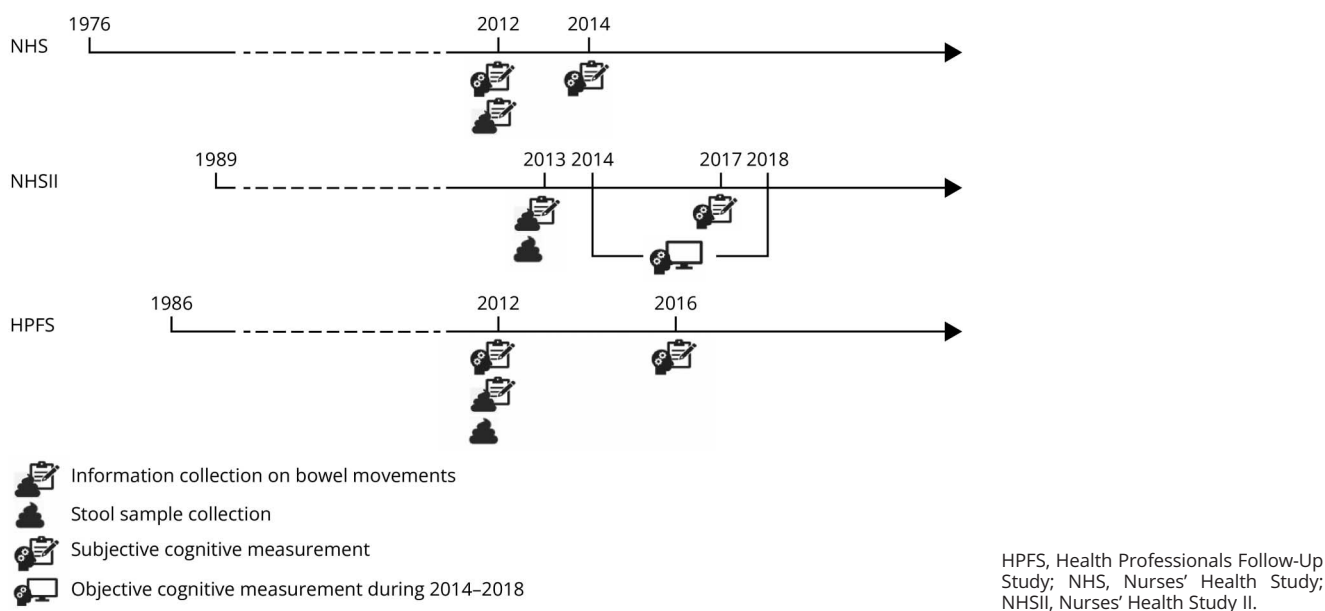
The NHSII is a prospective cohort study of 116,429 female registered nurses aged 25–42 years who completed a questionnaire on diet, demographics, lifestyle, medications, and medical history in 1989.¹² The NHS enrolled 121,700 female nurses aged 30–55 years in 1976, while the HPFS included 51,529 male health professionals aged 40–75 years in 1986. All 3 cohorts are followed by biennial questionnaires on medical history, lifestyle, diet, and newly diagnosed disease. Participants reported their bowel movement frequency in 2012 in NHS and HPFS and in 2013 in NHSII. Between 2014 and 2016, the cognitive function of 15,129 NHSII participants was assessed using the CogState neuropsychological battery. The subjective cognitive decline was assessed among NHSII participants in 2017, NHS participants in 2012 and 2014, and HPFS participants in 2012 and 2016. Figure 1 shows the timelines of measurements in the 3 cohorts.

The 3 cohorts have high follow-up rates, with a cumulative follow-up exceeding 85% of potential person time. Given that constipation has been considered a prodromal symptom of PD, we excluded PD at baseline and through the end of follow-up. We also excluded participants with a history of stroke or dementia at baseline and throughout the follow-up. In addition, we excluded participants who (1) did not report bowel movement frequency, (2) were younger than 50 years when they completed the CogState battery or the subjective cognitive function assessment, (3) failed integrity checks on all 4 CogState tasks, and (4) who had insufficient data to calculate at least one CogState composite cognitive score or one domain score of subjective cognitive function.

Standard Protocol Approvals, Registrations, and Patient Consents

The Institutional Review Boards of the Brigham and Women's Hospital and Harvard T.H. Chan School of Public Health approved the study protocol. Participants provided written informed consent for the collection of stool samples and the CogState neuropsychological battery. For questionnaire-based measurements, the participants' return of the questionnaire was

Figure 1 Overview of the Study on Bowel Movement Pattern, the Gut Microbiome, and Cognitive Function in Nurses' Health Study, Nurses' Health Study II, and Health Professionals Follow-Up Study



considered implied consent in accordance with the approval of the Institutional Review Boards.

Assessment of Bowel Movement Frequency and Covariates

NHS and NHSII participants reported their bowel movements with 6 possible answers: more than twice daily, twice daily, once daily, every other day, every 3–4 days, or every 5 days or less. HPFS participants responded to the same question with 4 possible answers: more than once daily, daily, every other day, or every 3 days or less. Demographic data, medical history, medication use, body weight and height, marital status, and living arrangement were collected at baseline and updated biennially through self-administered questionnaires in all 3 studies. Depression status was determined through the five-item version of the Mental Health Inventory (MHI-5) and self-reported physician's diagnosis of depression. In addition, participants in the 3 cohorts reported their use of antidepressant medication, antibiotics, and laxatives every 2–4 years. Participants' socioeconomic status (SES) was evaluated by using Census-aggregated characteristics of block groups of residence every 2 years since the beginning of the study. The participants' usual diet was assessed in the 3 cohorts every 2–4 years through validated, semiquantitative food frequency questionnaires (sFFQs), which asked about their consumption of foods, beverages, and supplements during the previous year.¹³ The nutrient intake data were obtained from the sFFQ and calculated based on the Harvard University Food Composition Database.¹⁴ We applied the Alternative Healthy Eating Index, a validated dietary score, to quantify the overall dietary quality.¹⁵ In the 3 cohorts, participants' physical activity level was evaluated using validated questionnaires every 2–4 years.¹⁶

Body mass index (BMI) was calculated from body height and weight. We collected genetic information from a subpopulation of 22,047 participants and determined their apolipoprotein E (APOE) genotype using a TaqMan assay (Applied Biosystems, Foster City, CA) or a GWAS chip.¹⁷

Cognitive Function Assessment

We used the CogState computerized neuropsychological battery to evaluate the objective cognitive function of NHSII participants.¹⁸ This battery comprised 4 tasks: detection, identification, one card learning, and one back. Details of the CogState battery and our methods for calculating task scores can be found elsewhere^{18,19} and in the eMethods. We calculated 2 composite scores: the psychomotor speed/attention composite (averaging detection and identification task Z-scores) and the learning/working memory composite (averaging one card learning and one back task Z-scores). An overall cognitive function score was obtained by averaging the Z-scores of the 2 composite scores. All scores were coded so that higher scores indicated better cognitive performance. The CogState battery has demonstrated good validity and high test-retest reliability^{18,20–22}; sensitivity in detecting subtle cognitive decline; ability to differentiate between normal cognitive function, MCI, and AD²⁰; and, in NHSII, strong association with APOE-ε4 allele carriage ($p = 0.001$).

Subjective cognitive decline was measured using a set of yes/no questions, consisting of 7 questions in NHSII and NHS and 6 questions in HPFS, on self-perceived recent changes in memory and cognition. Details of the assessment method were described elsewhere and in the eMethods.²³ In brief, a participant answered whether they had more trouble than usual in performing the activities that pertain to 4 domains of

cognitive function, including general memory, executive function, attention, and visuospatial skills. Our validation studies have shown strong and sensitive associations between subjective cognitive decline and objective cognitive function measured using a telephone-based neuropsychological battery and a strong association with APOE-ε4 allele carriage. For each question, 1 was assigned for a “yes” response and 0 for a “no” response. These values were then summed to calculate an overall score for subjective cognitive decline and its respective domains, with higher scores indicating worse cognitive function. The average score was computed for participants in NHS and HPFS who had repeated measurements. To evaluate the likelihood of experiencing more subjective cognitive complaints between the first and second assessments, a binary outcome was created. A value of 1 indicated that a participant reported more subjective cognitive complaints during the second assessment, while a value of 0 indicated otherwise. This binary outcome was calculated for NHS and HPFS participants who completed both assessments, which accounted for 86% of the participants.

Gut Microbiome Substudy

From 2011 to 2013, 520 NHSII and HPFS participants, free from coronary heart disease, stroke, cancer, or major neurologic disease, provided 1 or 2 pairs of self-collected stool samples. Details on stool sample collection and immediate *ex situ* conservation of metagenomic components, laboratory handling, and paired end shotgun sequencing of DNA can be found in our previous publications.²⁴ Whole-genome metagenomic data were obtained using the Illumina HiSeq paired end (2 × 101 nucleotides) shotgun sequencing platform. We generated taxonomic profiles by applying the MetaPhlAn 3.²⁵ We used MMUPHin²⁶ to correct the potential batch effect in the metagenomic data sets.

Statistical Analysis

We grouped the study population into 4 categories of bowel movement frequency: every 3 or more days, every other day, daily, twice, or more per day. We standardized the cognitive function scores by calculating cohort-specific Z-scores. General linear models were used to quantify the multivariable-adjusted differences and their 95% confidence intervals (CIs) in cognitive function score, comparing each category of bowel movement frequency with the reference category (once daily). We used Poisson regression to quantify the association between bowel movement frequency and subjective cognitive decline. We calculated ORs (95% CIs) for 3-unit increments in subjective cognitive decline based on our previous study that found that ≥3 positive answers to the subjective cognitive decline questions were an indicator of poor cognitive function.^{27,28} In addition, we calculated ORs and their 95% CIs of reporting more subjective cognitive complaints over time, comparing higher and lower categories with the daily bowel movement, using logistic regression models. The covariables included in all the models can be found in the footnotes of the tables. To examine the linear trend across categories of bowel movement frequency, we included the midpoint of each bowel movement frequency category as a scored trend variable, modeled the variable continuously, and used the Wald test to calculate *p* values for

linear trend. To quantify nonlinearity in the association, we used restricted cubic splines with 3 knots to flexibly model nonlinear associations.²⁹ *p* value for nonlinearity was calculated using a likelihood ratio test comparing 2 nested models with and without restricted cubic spline terms. We conducted a series of secondary analyses to test the robustness of our findings. First, we continuously modeled the subjective cognitive scores and quantified the association between bowel movement frequency and the scores using general linear models. Second, we examined the associations of bowel movement frequency with cognitive function and the risk of reporting more subjective cognitive complaints over time in subgroups defined by major confounding variables and risk factors for cognitive decline. Third, we examined the association between bowel movement frequency measured long before the cognitive assessments (1982 in NHS and 2000 in HPFS) and subjective cognitive decline to address the potential issue of reverse causation. Fourth, as fiber supplement use may indicate constipation, we conducted a sensitivity analysis by excluding fiber supplement users. Last, we quantified the prospective association between constipation (bowel movement every 3 days or less frequently) and the risk of a composite end point of self-reported dementia and deaths due to dementia during a 7-year follow-up (2012–2019) in NHS using Cox proportional hazard model.

In the microbiome analysis, we first applied permutational multivariate analysis of variance (PERMANOVA) to quantify the percentage of variance in microbial communities explained by bowel movement frequency, subjective cognitive decline, and covariables based on the Bray-Curtis (BC) dissimilarity metric (*n* = 999 permutations). Second, we performed dimension reduction using principal coordinate analysis based on the BC dissimilarity metric and calculated the first principal coordinate loading score (PCo1) to summarize the general structure of the microbial community. Third, we examined bowel movement frequency and subjective cognitive function associations with individual microbial features using the MaAsLin2,³⁰ a method designed for quantifying exposure-microbe associations based on linear mixed models. Last, because of the known impact of antibiotic or laxative use on bowel movement frequency, we performed sensitivity analyses among nonusers of antibiotics or laxatives. All the high-dimensional tests in our microbiome analysis were corrected for multiple hypothesis testing by controlling the false discovery rate using the Benjamini-Hochberg method with a target rate of 0.25 for *q* values. Details of microbiome analysis can be found in the eMethods (links.lww.com/WNL/D131). All the analyses were performed using SAS v 9.4 (SAS Institute, Cary, NC) or R version 4.1.2 (The R Foundation). All *p* values were 2-tailed (α = 0.05).

Data Availability

All the data from the Nurse's Health Study, the Nurses' Health Study II, and the Health Professionals Follow-Up Study are available through a request for external collaboration and on approval of a letter of intent and a research proposal. Details on how to request external collaborations with the Nurse's Health Study, the Nurses' Health Study II,

Table 1 Characteristics of the Study Population (n = 112,753) According to the Frequency of Bowel Movements in the Nurses' Health Study, the Nurses' Health Study II, and the Health Professionals Follow-Up Study at Baseline

	Nurses' Health Study (2012)				Nurses' Health Study II (2013)				Health Professionals Follow-Up Study (2012)			
	Every 3+ d	Every 2 d	Daily	≥Twice/day	Every 3+ d	Every 2 d	Daily	≥Twice/day	Every 3+ d	Every 2 d	Daily	≥Twice/day
n	2,473	6,153	24,959	9,165	3,647	8,940	30,494	11,807	301	1,063	9,076	4,675
Age, y	76.5	76.2	75.7	74.9	58.2	58.2	58.6	58.9	78.6	77.1	74.7	73.3
Body mass index, kg/m ²	25.9	26.1	26.2	26.6	26.7	26.8	27.2	28.7	25.2	25.1	25.2	25.2
Alternate Healthy Eating Index scores	59.3	60.3	62.1	62.5	62.5	64.0	65.4	65.2	63.7	66.5	67.9	68.4
Fiber intake, g/d	19.0	20.1	21.0	21.9	20.9	21.8	22.8	23.5	22.3	24.0	25.4	26.8
Physical activity, METs/wk	25.1	26.4	28.6	28.1	24.2	26.8	30.3	28.6	27.8	32.4	39.1	41.3
Neighborhood SES index ^a	−0.4	−0.1	0	−0.1	−0.1	0.1	0.2	−0.1	0.2	0	0	0
Married, %	60.0	59.7	59.2	58.8	77.3	76.7	76.1	74.9	84.8	84.0	83.9	85.1
Living arrangement, alone, %	28.1	28.5	29.8	30.0	10.7	11.4	11.9	12.2	10.7	11.0	11.3	9.9
Hypercholesterolemia, %	67.3	66.1	63.5	64.0	52.4	49.2	47.1	51.4	61.0	58.3	55.5	54.5
Hypertension, %	64.7	64.9	62.9	66.1	35.0	33.5	34.0	40.1	55.7	54.3	55.0	55.7
Diabetes, %	14.8	14.7	12.0	13.7	7.8	7.1	6.6	9.7	14.8	16.4	11.8	11.7
Hyperthyroidism ^b , %	n/a	n/a	n/a	n/a	3.3	2.8	2.8	3.0	n/a	n/a	n/a	n/a
Hypothyroidism, %	22.2	20.1	20.3	22.3	24.3	20.8	20.0	22.8	n/a	n/a	n/a	n/a
Depression & Antidepressant use ^b , %	28.5	22.2	16.7	19.9	43.9	34.3	28.7	35.5	17.8	15.5	10.7	12.5
Antibiotic use, %	71.6	72.7	70.3	74.3	73.5	70.8	70.1	72.5	43.0	48.2	46.7	47.6
Laxative use, %												
Never	30.5	45.9	66.8	73.3	39.4	55.7	72.0	78.4	43.7	52.4	69.9	76.2
Once/month - once/week	31.1	25.5	13.0	7.9	33.9	25.3	15.1	9.6	31.6	26.8	15.9	10.2
Twice+/week	38.3	28.6	20.3	18.8	26.6	19.0	13.0	12.0	24.7	20.8	14.2	13.6
Smoking status, %												
Never	45.0	46.1	46.7	48.1	64.6	67.0	66.1	66.9	34.0	39.0	38.9	40.5
Past	50.4	48.6	48.6	48.0	30.0	29.3	29.7	28.7	59.9	57.8	58.4	57.5
Current	4.4	5.1	4.5	3.7	5.4	3.7	4.2	4.4	6.0	3.1	2.7	2.0
Family history of dementia ^c , %	21.3	21.2	21.7	23.4	n/a	n/a	n/a	n/a	27.8	24.7	25.0	26.2
Spouse's education level ^c , %												
High school or less	35.7	34.0	32.0	32.4	17.8	15.8	14.8	15.1	n/a	n/a	n/a	n/a
College degree	22.4	22.8	22.9	23.1	43.8	43.8	43.4	43.8	n/a	n/a	n/a	n/a
Graduate school	18.0	17.3	20.5	20.1	24.4	26.8	28.0	26.6	n/a	n/a	n/a	n/a
Unknown	23.9	25.9	24.6	24.4	14.0	13.6	13.9	14.5	n/a	n/a	n/a	n/a

Values are means for continuous variables and percentages for categorical variables.

^a Neighborhood SES, neighborhood socioeconomic status, index as Z-scores.

^b Physician-diagnosed depression and self-reported antidepressant use.

^c n/a means that the variable was not measured in the study population.

and the Health Professionals Follow-Up Study can be found at sites.sph.harvard.edu/hpfs/for-collaborators/ and nurseshealthstudy.org/researchers.

Results

Population Characteristics

The final analytical population consisted of 12,696 women in the analysis of objective cognitive function and 112,753 women and men in the analysis of subjective cognitive decline (Table 1). The mean age of the study population was 75.7 years in NHS, 58.6 in NHSII, and 74.5 in HPFS at the time of bowel movement frequency assessment. Participants who reported more frequent bowel movements had a higher dietary fiber intake but used laxatives less frequently (Table 1). Compared with those with a bowel movement frequency of once daily, participants with more and less frequent bowel movements were less physically active and more likely to have depressive symptoms and use antidepressants.

Objective Cognitive Function

After adjustment for potential confounders, bowel movement frequency was associated with overall cognitive function measured by the CogState battery in an inverse J-shape dose-response manner ($p_{\text{nonlinearity}} = 0.04$, Table 2). We

found similar dose-response associations for learning and working memory ($p_{\text{nonlinearity}} = 0.01$) and psychomotor speed and attention, although the association was not statistically significant for the latter ($p_{\text{nonlinearity}} = 0.58$). Compared with those with daily bowel movement, the magnitude of cognitive worsening in participants with bowel movement frequency of every 3 days or less was equivalent to 3.0 years of aging. For learning and working memory, participants with bowel movement frequency of every 3 days or less had a cognitive worsening equivalent to 7.1 years of aging.

Subjective Cognitive Decline

We found inverse J-shape dose-response associations between bowel movement frequency and subjective cognitive decline in each of the 3 independent cohorts (all $p_{\text{nonlinearity}} < 0.001$, Table 3). In our pooled analysis, the lowest (every 3 days or less) and highest (twice a day or more) categories of bowel movement, compared with daily bowel movement, were associated with 73% (OR, 1.73; 95% CI 1.60, 1.86) and 37% (OR, 1.37; 95% CI 1.33, 1.44) higher odds of subjective cognitive decline, respectively. In addition, when we modeled the overall subjective cognitive decline and different domains, including general memory, executive function, attention, and visuospatial skills, as continuous scores, the bowel movement frequency was associated with the scores in a J-shape dose-response manner (all $p_{\text{nonlinearity}} < 0.001$, eTables 1 and 2, links.lww.com/WNL/D131).

Table 2 Association Between the Frequency of Bowel Movements (Reported in 2013) and Objective Cognitive Function (Measured in 2014–2018), Expressed in Years of Cognitive Aging (and 95% CI) in the Nurses' Health Study II^a

	Frequency of bowel movements				$p_{\text{nonlinearity}}^b$
	Every 3+ d	Every 2 d	Daily	≥Twice/day	
Overall cognitive function					
Model 1	3.5 (1.8 to 5.2)	−0.3 (−1.5 to 0.9)	0 (ref.)	0.8 (−0.3 to 1.8)	0.01
Model 2	3.0 (1.2 to 4.8)	−0.5 (−1.7 to 0.7)	0 (ref.)	0.7 (−0.3 to 1.8)	0.03
Model 3	3.0 (1.2 to 4.7)	−0.4 (−1.6 to 0.8)	0 (ref.)	0.4 (−0.7 to 1.4)	0.04
Learning and working memory					
Model 1	8.1 (3.6 to 12.5)	1.2 (−1.8 to 4.2)	0 (ref.)	3.6 (0.9 to 6.4)	0.001
Model 2	7.3 (2.8 to 11.8)	0.9 (−2.1 to 4.0)	0 (ref.)	3.3 (0.6 to 6.0)	0.005
Model 3	7.1 (2.5 to 11.6)	0.9 (−2.2 to 3.9)	0 (ref.)	2.5 (−0.3 to 5.2)	0.01
Psychomotor speed and attention					
Model 1	2.0 (0.2 to 3.8)	−0.7 (−1.9 to 0.5)	0 (ref.)	−0.1 (−1.2 to 1.0)	0.36
Model 2	1.6 (−0.2 to 3.4)	−0.9 (−2.1 to 0.3)	0 (ref.)	−0.1 (−1.2 to 1.0)	0.63
Model 3	1.7 (−0.2 to 3.5)	−0.8 (−2.0 to 0.4)	0 (ref.)	−0.3 (−1.4 to 0.8)	0.58

Model 1 adjusted for age at cognitive function assessment and educational attainment of parents and spouse.

Model 2 further adjusted for antibiotic use, laxative use, antidepressant use, and symptoms of depression.

Model 3 further adjusted for neighborhood socioeconomic status index, Alternative Healthy Eating Index, fiber intake, smoking status, physical activity level, marriage status, living arrangement, and histories of hypertension, diabetes, hypercholesterolemia, hyperthyroidism, and hypothyroidism.

One year of aging was associated with decreases of 0.02 in the overall cognitive function score, 0.01 in the score for learning and working memory, and 0.04 in the score for psychomotor speed and attention. We standardized the cognitive scores to Z-scores.

^a Values are multivariable-adjusted mean differences (95% CI) expressed in years of cognitive aging in cognitive function comparing participants with higher or lower frequencies of bowel movement to those with daily bowel movement estimated from general linear models.

^b We modeled nonlinear associations between bowel movement frequency and cognitive function scores using restricted cubic splines; potential nonlinearity was tested by using a likelihood ratio test.

Table 3 Association Between Bowel Movement Frequency and 3-Unit Increments in Subjective Cognitive Decline in the Nurses' Health Study (NHS), the Nurses' Health Study II (NHSII), and the Health Professionals Follow-Up Study (HPFS)^a

	Frequency of bowel movement				$p_{\text{nonlinearity}}^b$
	Every 3+ d	Every 2 d	Daily	≥Twice/day	
NHS					
Model 1	2.70 (2.42–3.01)	1.62 (1.49–1.76)	1 (ref.)	1.41 (1.31–1.51)	<0.001
Model 2	1.99 (1.78–2.22)	1.37 (1.26–1.48)	1 (ref.)	1.38 (1.28–1.49)	<0.001
Model 3	1.90 (1.61–2.01)	1.30 (1.20–1.41)	1 (ref.)	1.41 (1.31–1.51)	<0.001
NHSII					
Model 1	2.64 (2.39–2.92)	1.54 (1.43–1.65)	1 (ref.)	1.53 (1.43–1.64)	<0.001
Model 2	1.84 (1.66–2.03)	1.32 (1.22–1.42)	1 (ref.)	1.40 (1.31–1.50)	<0.001
Model 3	1.64 (1.48–1.82)	1.27 (1.18–1.37)	1 (ref.)	1.36 (1.27–1.46)	<0.001
HPFS					
Model 1	2.49 (1.78–3.49)	1.37 (1.11–1.70)	1 (ref.)	1.25 (1.10–1.42)	<0.001
Model 2	1.94 (1.38–2.72)	1.18 (0.96–1.47)	1 (ref.)	1.27 (1.12–1.45)	<0.001
Model 3	1.61 (1.13–2.28)	1.06 (0.85–1.32)	1 (ref.)	1.32 (1.16–1.51)	<0.001
Pooled ^c					
Model 1	2.69 (2.46–2.86)	1.56 (1.48–1.64)	1 (ref.)	1.44 (1.37–1.52)	<0.001
Model 2	1.91 (1.77–2.05)	1.33 (1.26–1.40)	1 (ref.)	1.37 (1.33–1.44)	<0.001
Model 3	1.73 (1.60–1.86)	1.26 (1.19–1.33)	1 (ref.)	1.37 (1.33–1.44)	<0.001

Model 1 adjusted for age at cognitive function assessment and educational attainment of parents and spouse.

Model 2 further adjusted for antibiotic use, laxative use, antidepressant use, and symptoms of depression.

Model 3 further adjusted for neighborhood socioeconomic status index, Alternative Healthy Eating Index, fiber intake, smoking status, physical activity level, marriage status, living arrangement, and histories of hypertension, diabetes, hypercholesterolemia, hyperthyroidism, and hypothyroidism.

^a Values are multivariable-adjusted odds ratios (95% CI) of 3-unit increments in subjective cognitive decline comparing participants with higher or lower frequencies of bowel movement to those with daily bowel movement estimated from Poisson regression models.

^b We modeled nonlinear associations between bowel movement frequency and cognitive function scores using restricted cubic splines; potential nonlinearity was tested by using a likelihood ratio test.

^c Results for NHS, NHSII, and HPFS from the multivariable model were combined using the fixed effects model.

We also found that extremes of bowel movement frequency were associated with increased subjective cognitive complaints over time, with repeated assessments ($p_{\text{nonlinearity}} < 0.001$, Table 4). Compared with those with a bowel movement frequency of once daily, participants with bowel movement frequencies of every 3 days or less and twice a day or more had 19% (OR = 1.19; 95% CI 1.07, 1.33) and 10% higher (OR = 1.10; 95% CI 1.04, 1.17) likelihood of more subjective cognitive complaints over time, respectively. Similarly, we found nonlinear dose-response relationships between bowel movement frequency and the likelihood of overtime worsening in general memory ($p_{\text{nonlinearity}} = 0.06$), executive function ($p_{\text{nonlinearity}} = 0.01$), attention ($p_{\text{nonlinearity}} < 0.001$), and visuospatial skills ($p_{\text{nonlinearity}} = 0.06$, eTable 3, links.lww.com/WNL/D131).

Subgroup and Secondary Analyses

In the subgroup analyses (eTables 4–6, links.lww.com/WNL/D131), the associations of bowel movement frequency with objective cognitive function were generally consistent across subgroups defined by age, laxative use, antibiotic use, histories

of hypertension, hypercholesterolemia, diabetes, smoking status, weight status, dietary quality, fiber intake, and APOE genotype (all $p_{\text{interaction}} > 0.05$). However, the associations of bowel movement frequency with subjective cognitive decline and more subjective cognitive complaints over time were more pronounced in APOE ε4 allele noncarriers and participants without hypercholesterolemia, diabetes, or depression. In addition, we found greater magnitudes of associations for bowel movement frequency comparing every 3 days or less vs daily with subjective cognitive decline in antibiotic users ($p_{\text{interaction}} = 0.02$) and with more subjective cognitive complaints over time in laxative users ($p_{\text{interaction}} < 0.001$).

The results for objective cognitive function and subjective cognitive decline did not change materially in a sensitivity analysis that excluded fiber supplement users (eTable 7, links.lww.com/WNL/D131). In another sensitivity analysis that leveraged the bowel movement frequency measured long before the cognitive assessments (1982 in NHS and 2000 in HPFS), we observed a similar albeit attenuated inverse

Table 4 Association Between Frequency of Bowel Movements and the Likelihood of More Subjective Cognitive Complaints Over Time in the Nurses' Health Study (NHS) and the Health Professionals Follow-Up Study (HPFS)^a

	Frequency of bowel movements				<i>p</i> _{nonlinearity} ^b
	Every 3+ d	Every 2 d	Daily	≥Twice/day	
NHS					
Cognitive decline/total	446/2,090	970/5,342	3,608/22,250	1,459/8,311	
Model 1	1.37 (1.23–1.53)	1.13 (1.04–1.22)	1 (ref.)	1.14 (1.06–1.22)	<0.001
Model 2	1.25 (1.11–1.40)	1.07 (0.99–1.16)	1 (ref.)	1.13 (1.06–1.21)	<0.001
Model 3	1.20 (1.07–1.35)	1.05 (0.97–1.14)	1 (ref.)	1.14 (1.06–1.22)	<0.001
HPFS					
Cognitive decline/total	62/200	229/766	1,860/7,415	941/3,875	
Model 1	1.21 (0.88–1.65)	1.22 (1.03–1.44)	1 (ref.)	1.02 (0.93–1.12)	0.02
Model 2	1.12 (0.82–1.53)	1.16 (0.98–1.38)	1 (ref.)	1.03 (0.94–1.13)	0.08
Model 3	1.10 (0.80–1.52)	1.11 (0.94–1.32)	1 (ref.)	1.04 (0.94–1.14)	0.17
Pooled ^c					
Model 1	1.35 (1.22–1.50)	1.14 (1.06–1.23)	1 (ref.)	1.10 (1.04–1.16)	<0.001
Model 2	1.23 (1.11–1.37)	1.09 (1.01–1.17)	1 (ref.)	1.10 (1.04–1.16)	<0.001
Model 3	1.19 (1.07–1.33)	1.06 (0.98–1.14)	1 (ref.)	1.10 (1.04–1.17)	<0.001

Model 1 adjusted for age at cognitive function assessment and educational attainment of parents and spouse.

Model 2 further adjusted for antibiotic use, laxative use, antidepressant use, and symptoms of depression.

Model 3 further adjusted for neighborhood socioeconomic status index, Alternative Healthy Eating Index, fiber intake, smoking status, physical activity level, marriage status, living arrangement, and histories of hypertension, diabetes, hypercholesterolemia, hyperthyroidism, and hypothyroidism.

^a Values are multivariable-adjusted odds ratios (95% CI) of having more subjective cognitive complaints over time comparing participants with higher or lower frequencies of bowel movement to those with daily bowel movement estimated from logistic regression models. Follow-up period was 2012–2014 for NHS and 2012–2016 for HPFS.

^b We modeled nonlinear associations between bowel movement frequency and cognitive function scores using restricted cubic splines; potential nonlinearity was tested by using a likelihood ratio test.

^c Results for NHS and HPFS from the multivariable model were combined using the fixed effects model.

J-shaped dose-response pattern in the relationship between bowel movement frequency and the ORs of subjective cognitive decline ($p_{\text{nonlinearity}} < 0.001$, eTable 7).

During a 7-year follow-up in NHS, we documented 1,629 self-reported dementias or deaths due to dementia. Compared with those with more frequent bowel movements, participants with bowel movement frequency of every 3 days or less had a 19% higher risk of dementia (hazard ratio = 1.19, 95% CI 1.00–1.42, $p = 0.06$).

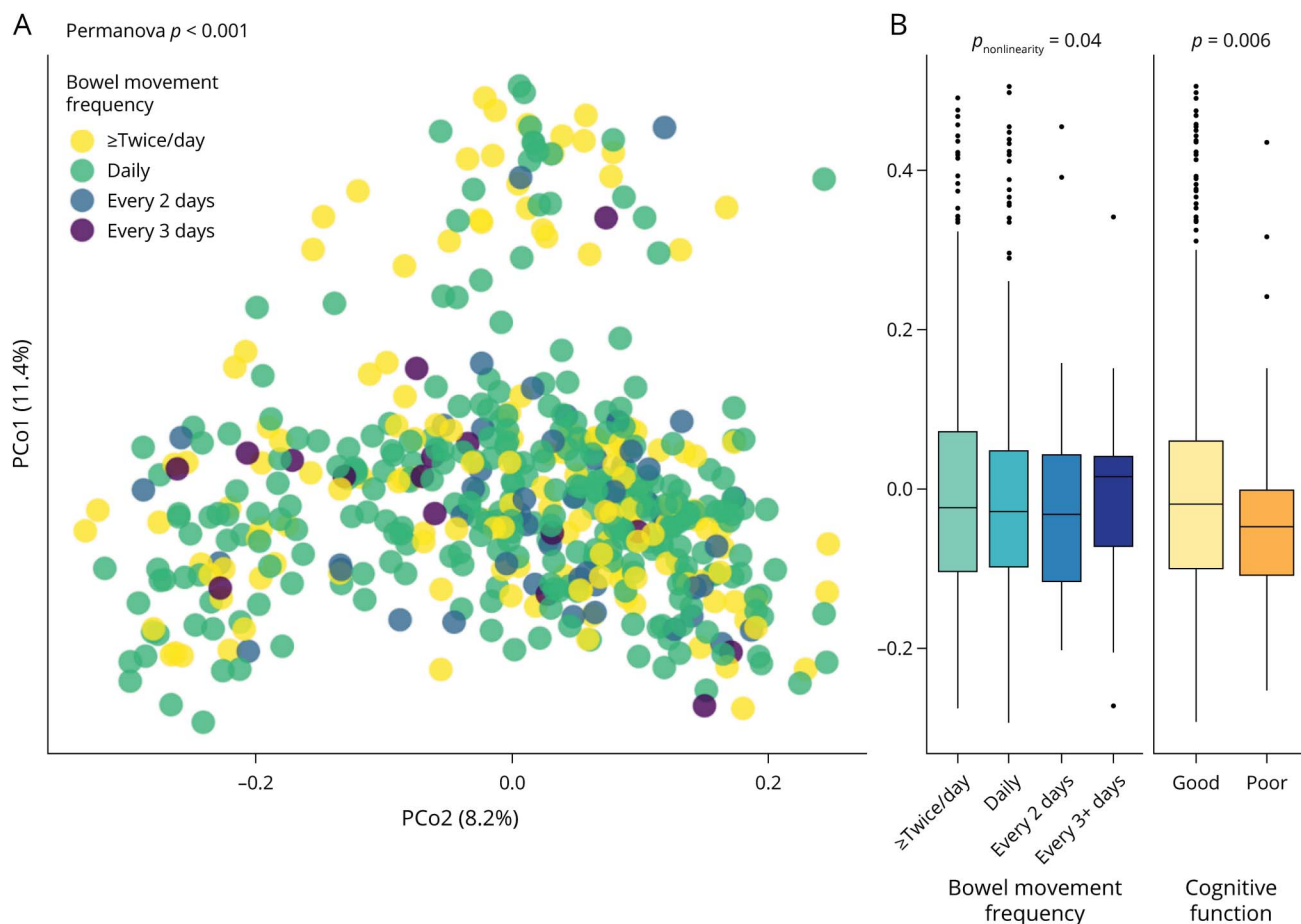
Gut Microbiome Substudy

Our study included 515 participants who had data on bowel movement frequency, gut microbiome, and subjective cognitive decline. Although bowel movement frequency was not a major driver of the overall gut microbial composition (Figure 2), PERMANOVA revealed that its association was significant with the overall variation of the gut microbiome ($p < 0.001$, eFigure 1, links.lww.com/WNL/D131). There was a significant nonlinear association between bowel movement frequency and PCo1 ($p_{\text{nonlinearity}} = 0.04$). In addition, subjective cognitive decline was significantly associated with the overall variation of the gut microbiome ($p = 0.004$) and

positively associated with PCo1 ($p_{\text{trend}} = 0.006$). Furthermore, bowel movement frequency explained the highest percentages of variation (1.15%) in the gut microbiome among a group of variables that included antibiotic use, age, sex, overall dietary quality, fiber intake, laxative use, and Bristol Stool Scale (eFigure 1). The percentages of variation explained by bowel movement frequency (1.2%) were higher than those explained by antibiotic use (0.6%) and the Bristol stool scale (0.4%), 2 covariables previously reported to have strong influences on the gut microbiome.^{31,32}

We identified 19 microbial species significantly associated with bowel movement frequency and 15 species for cognitive function; 3 species were significantly associated with both bowel movement frequency and cognitive function, including *Ruminococcus gnavus*, *Clostridium citroniae*, and *Anaeromassili-bacillus* sp An250 (Figure 3). Dysbiosis-related *Veillonella* spp., such as *V. parvula*, *V. dispar*, and *V. atypica*, gram-negative and strictly anaerobic bacteria commonly found in the oral cavity and the intestinal and genitourinary tract,³³ were associated with more frequent bowel movements. A reduction of the abundance of *Roseburia* spp., including *R. inulinivorans* and *R. faecis*, the predominant butyrate-producing bacterial species,³⁴

Figure 2 Associations of Bowel Movement Frequency and Cognitive Function With Overall Variation of the Gut Microbiome



The scatter plot (A) presents results from the principal coordinate analysis based on Bray-Curtis dissimilarity calculated from species-level taxonomy data in 515 participants from Nurse' Health Study II and Health Professionals Follow-Up Study. The significance level for the association between bowel movement frequency and the overall configuration of the gut microbial composition was calculated based on permutational multivariate analysis of variance (PERMANOVA) with 999 permutations. The boxplots (B) show the association between bowel movement frequency and the 1st principal coordinate score (PCo1) and the association between PCo1 and categories of cognitive function using multivariable linear regression adjusted for age, antibiotic use, laxative use, Bristol stool scale, overall dietary quality, and fiber intake. Based on the overall cognitive function score, we created 2 categories of cognitive function, including "poor" ($3 \leq$ overall score ≤ 7) and "good" ($0 \leq$ overall score < 3). The test of nonlinear dose-response association between bowel movement frequency and PCo1 was performed using restricted cubic spline regression based on the likelihood ratio test.

was observed in participants with less frequent bowel movements. For cognitive function, enrichment of *Erysipelatoclostridium ramosum*, one of the opportunistic pathogens,³⁵ and *Blautia* sp. CAG 257, an anaerobic bacterium, was associated with worse cognitive function. *Eubacterium eligens*, a dietary fiber processor and butyrate producer, was enriched in participants with better cognitive function. The results did not change materially after excluding antibiotic or laxative users (eFigure 2, links.lww.com/WNL/D131).

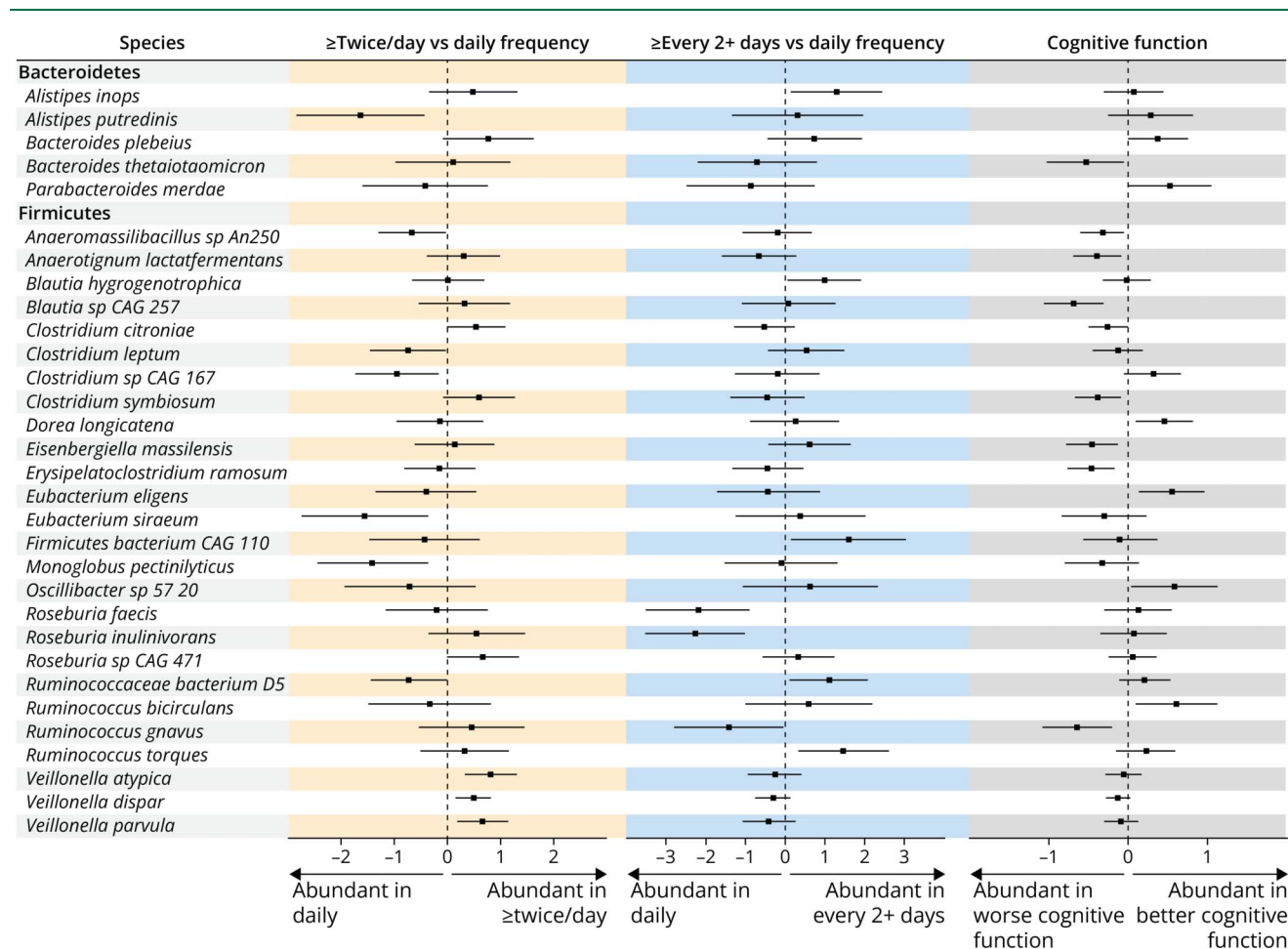
Discussion

In 3 large cohort studies of US men and women, we found significant, inverse J-shape dose-response associations of bowel movement frequency with objectively measured cognitive function scores and the risk of subjective cognitive decline. Participants with less frequent bowel movements had

a worse objectively measured cognitive function and a higher risk of subjective cognitive decline. We also found suggestive evidence that constipation was prospectively associated with a higher risk of dementia. Our exploratory analysis in a subpopulation found that the enrichment of dysbiosis-associated, proinflammatory microbial species and depletion of butyrate-producing microorganisms may partially explain the mechanistic link between abnormal intestinal motility patterns and cognitive function.

Data on variations in intestinal motility, cognitive function, and risk of dementia are sparse. In 751 community-dwelling participants in Singapore, Huang et al.⁴ reported a cross-sectional association between bowel movement frequency and MCI, in which bowel movement of 4 times per week or more, compared with 3 times per week or less, was associated with an OR of MCI of 0.58 (95% CI 0.36–0.94). Another cross-sectional study conducted in China found a higher

Figure 3 Associations of Bowel Movement Frequency and Cognitive Function With Species-Level Gut Microbial Features ($q < 0.25$)



Values are beta coefficients (95% CI) derived from multivariable-adjusted linear mixed models linking bowel movement frequency or cognitive function (Z-score) with relative abundance of microbial species. The linear mixed models simultaneously adjusted for age, sex, antibiotic use, laxative use, Bristol stool scale, overall dietary quality, and fiber intake. This analysis was based on 1,688 stool samples from 515 participants from Nurse' Health Study II and Health Professionals Follow-Up Study. The widths of confidence intervals were adjusted for multiple comparison using the Benjamini-Hochberg method with a target false discovery rate of 0.25.

prevalence of constipation among individuals with dementia and MCI compared with cognitively healthy individuals.³⁶ Our prospective findings on worse cognitive function and higher risk of cognitive decline in individuals with less frequent bowel movements are consistent with the existing evidence. More recently, a prospective study from the UK Biobank linked laxative use, a potential surrogate of constipation, with a 51% higher risk of dementia (hazard ratio = 1.51; 95% CI 1.30–1.75), lending further support to our findings.⁹ However, the Singaporean and Chinese studies measured bowel movement frequency as a dichotomous variable, precluding an examination of the dose-response relationship. In addition, our study provided first-of-its-kind evidence that examined a wide spectrum of bowel movement frequency, especially an analysis of the more frequent end, in relation to cognitive function. Our subgroup analysis found more pronounced associations of bowel movement frequency with subjective cognitive decline and more subjective cognitive complaints over time in APOE ε4 allele noncarriers. It is

possible that the increased risk of cognitive decline in APOE ε4 allele carriers is primarily driven by their genetic predisposition, with the limited population-level variation likely to be explained by nongenetic risk factors. Furthermore, this finding aligns with emerging research suggesting that APOE plays a role in modulating the gut microbiome and that the gut microbiome, in turn, regulates neuroinflammation and tau-related neurodegeneration in an APOE isoform-specific manner.^{37–39}

We found distinct patterns in the gut microbial community across categories of bowel movement frequencies. Abnormally high abundance of oral-type, proinflammatory bacteria, for example, *V. dispar* and *V. atypica*, in the gut, potentially indicating a tendency toward dysbiosis, a systemic perturbation throughout the digestive tract due to the toxic and mutagenic metabolites produced by oral bacteria,⁴⁰ was found in participants with more frequent bowel movements. We found lower abundance of *R. inulinivorans* and *R. faecis* in those with less frequent bowel movements. Depleted *Roseburia* spp. is

one of the most consistently reported alternations in the gut microbiome studies in PD.⁴¹ *Roseburia* constitutes a major group of butyrate-producing Firmicutes that go through the butyryl-CoA:acetate CoA-transferase route for butyrate synthesis.⁴² In vivo and in vitro studies have shown that butyrate interacts closely with the immune system by activating G-protein-coupled receptors and inhibiting the activity of histone deacetylases, which results in a reduction of neuro-inflammation and oxidative stress.⁴³⁻⁴⁵ In addition, butyrate has been proven to improve the brain-blood barrier permeability and its protective function by tight junction.⁴² Therefore, the observed reduction of *Roseburia* spp. in participants with low bowel movement frequency may explain the inverse J-shape association between bowel movement frequency and cognitive function.

Our findings also support a gut microbial contribution to cognitive impairment. We found enrichment of *Erysipelato-clostridium ramosum* (previously known as *Clostridium ramosum*) and *Blautia* sp. CAG 257 was associated with worse cognitive function. Consistent with previous studies on the gut microbiome of individuals with neurologic disorders, a higher abundance of *C. ramosum* and *Blautia* spp. was observed in patients with AD and children with neuro-developmental disorders.⁴⁶ We also found proinflammatory bacterium, *Ruminococcus gnavus*, was enriched in the gut of participants with worse cognitive function. By contrast, *Eubacterium eligens*, one of the plant-derived polysaccharide processors and butyrate producers, was enriched in participants with better cognitive function.

Our study has several limitations. First, because our study was observational in nature, causality cannot be established. Second, residual confounding could not be ruled out, although we adjusted for many potential confounders. Our findings of significant interactions between bowel movement frequency and antibiotics, laxatives, diabetes, or depression suggested that medications and disease status may be strong confounding factors. However, in our subgroup analyses, the observed nonlinear dose-response relationship remained significant but differed only in magnitude across subgroups defined by medication use and disease status. Third, we examined the subjective cognitive decline during a relatively short period. Therefore, it is important to acknowledge that our findings may be susceptible to bias due to reverse causality. In addition, owing to the limited duration of follow-up, we were unable to investigate the longer-term temporality and induction period. However, it is worth noting that our sensitivity analysis, which incorporated bowel movement frequency data measured much earlier (1982 in NHS and 2000 in HPFS), demonstrated similar associations. This partially addresses the aforementioned concern. Last, we only collected data on the gut microbiome at baseline and thus could not examine the impact of changes in the gut microbiome on cognitive decline. The strengths of this study included large sample size, the inclusion of 3 independent cohorts to ensure reproducibility of findings, comprehensive data collection on many important confounding factors,

complementary measurements of cognitive function, and a mechanistic study on the gut microbiome that uses whole-genome shotgun metagenomics.

In summary, less frequent bowel movements were associated with worse cognitive function. Perturbations in the gut microbiome may partially explain the mechanistic link between variations in intestinal motility patterns and cognitive function. Additional prospective studies are required to confirm the associations between bowel movement frequency and cognitive function and further elucidate the mechanistic role of the gut microbiome in cognitive impairment and dementia.

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Disclosure

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Francine Grodstein, PhD	Rush Alzheimer's Disease Center, Rush University Medical Center	Major role in the acquisition of data
Alberto Ascherio, MD, DrPH	Department of Nutrition, Harvard T.H. Chan School of Public Health	Major role in the acquisition of data

Appendix (continued)

Name	Location	Contribution
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Andrew T. Chan, MD, MPH	Division of Gastroenterology, Massachusetts General Hospital and Harvard Medical School	Major role in the acquisition of data
Curtis Huttenhower, PhD	Department of Biostatistics, Harvard T.H. School of Public Health	Major role in the acquisition of data
Meir J. Stampfer, MD, DrPH	Department of Epidemiology, Harvard T.H. Chan School of Public Health	Major role in the acquisition of data; study concept or design
Dong D. Wang, MD, ScD	Channing Division of Network Medicine, Department of Medicine, Brigham and Women's Hospital and Harvard Medical School	Major role in the acquisition of data; study concept or design; analysis or interpretation of data

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